Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/27/2024
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Cognitive and Physiologic Responses in Commercial Space Crew on Short-Duration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	12/01/2021	**End Date:**	03/31/2028
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	End date changed to 03/31/2028 per TRISH (Ed., 3/4/25). End date changed to 03/31/2025 per TRISH (Ed., 12/23/24). End date changed to 11/30/2024 per TRISH (Ed., 10/26/23). Previous end date was 11/30/2023.		
Key Personnel Changes/Previous PI:	Christopher Jones, Ph.D. added as CoI 2024		
COI Name (Institution):	Jones, Christopher Ph.D. (University of Pennsylvania Health System)		
Grant/Contract No.:	NNX16AO69A-EXP0009		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	October 2023 Update: The project was amended to collect data on SpaceX Polaris Dawn mission before, during and after a 5-day mission that includes an extravehicular activity (EVA), and Axiom's Ax-2 mission that will launch four spaceflight participants onboard a SpaceX Crew Dragon vehicle from Cape Canaveral, FL. This mission will dock to the International Space Station (ISS) United States Operating Segment (USOS). Once on orbit, the crewmembers will stay for approximately two weeks. The proposed research will be performed on two commercial crew flights: (1) The first one is performed by Space Adventures and includes one Japanese crew member who will launch on a Soyuz rocket from Baikonur in December 2021. He will spend a few days on the International Space Station (ISS) before his return to Russia. Data collection will be restricted to pre- and post-flight. (2) The second one is performed by Axiom Space and includes a crew of 4 (including one former female astronaut) who will launch on a Falcon rocket from Kennedy Space Center in December 2022. They will spend a few days on the International Space Station before they return to Florida. Data collection will be restricted to pre- and post-flight, but may include tests while docked to the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The primary objective of this study is to collect data on crew behavior and performance before, during (Polaris Dawn only) and after the mission, to extract the data, clean and annotate the data, and submit the data to TRISH's database. Lessons learned and descriptive data analyses will be presented at the HRP IWS in 2022 and 2023 (travel funding obtained through NASA grants) and at the 2023 ASMA meeting. The long-term goal of this TRISH effort is to generate a quickly growing commercial crew database that can be mined by the space research community. We facilitated a reliance agreement between UPenn and BCM IRB for SA and Ax-1. At this time, the Space Adventures mission, Ax-1 and Ax-2 were successfully completed. We are still waiting to download Ax-2 post-flight data from the iPad and Garmin watches, but otherwise all data were transmitted to the TrialX database. We facilitated another reliance agreement between UPenn and BCM IRB for Polaris Dawn that will likely translate to other future similar missions. We have extensively tested equipment (i.e., BioButton and Garmin smartwatch) and developed crew procedures for the Polaris Dawn mission. This includes generating NDAs with Garmin and BioIntelliSense. The hardware is currently undergoing bake-out procedures at SpaceX. We have already collected three pre-flight Cognition test bouts (plus survey) in all 4 PD crew. Commercial crew data have been presented at HRP IWS 2023 and the 2023 AsMA meeting. Another abstract for the 2024 HRP IWS has been submitted. The PI was invited as a keynote speaker to Garmin's 2023 Health Summit in Singapore to present on the Polaris Dawn mission and how we use smart sensors. An article on the Inspiration4 mission has been submitted to Nature and is currently under re-review.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Jones CW, Overbey EG, Lacombe J, Ecker AJ, Meydan C, Ryon K, Tierney B, Damle N, MacKay M, Afshin EE, Foox J. "Molecular and physiological changes in the SpaceX Inspiration4 civilian crew." Nature. 2024 Aug 29;632(8027):1155-64. https://doi.org/10.1038/s41586-024-07648-x ; PMID: 38862026 ; PMCID: PMC11357997 , Aug-2024